• POOR ENGINE IDLE QUALITY—VEHICLES EQUIPPED WITH 3.0L ENGINE ONLY—VEHICLES BUILT BEFORE 11/26/2001 ONLY • VIBRATION—ENGINE VIBRATION AT IDLE—VEHICLES EQUIPPED WITH 3.0L ENGINE—VEHICLES BUILT BEFORE 11/26/2001 ONLY	**Article No. 02-22-4**

FORD: 2002 RANGER

ISSUE

Some vehicles equipped with a 3.0L engine may exhibit a rough idle or engine vibration condition. This may be caused by the engine brackets and/or engine mounts and transmission insulator/retainer.

ACTION

Replace the current engine support brackets and/or engine mounts and transmission insulator/retainer with new parts. Refer to the following Service Procedure and parts table for details.

SERVICE PROCEDURE

For All Vehicles:

1. Verify that the engine is operating (idle) correctly by performing a cylinder balance test with WDS.
2. If idle quality is poor, with no driveability indicators, perform the following for each appropriate vehicle:
3. Disconnect the battery ground cable. Refer to Section 414-01.
4. Remove the hood.
5. Remove the fan. Refer to Section 303-03.
6. Attach the Engine Lifting Eyes.
7. Remove the engine support insulator nuts (Figure 1).
8. Raise the engine and support with a three-bar engine support.
9. Raise vehicle on a hoist.

NOTE

STEPS 10 THROUGH 14 CAN BE PERFORMED THROUGH THE FENDER WELL OPENINGS.

10. Remove right hand engine support bracket.

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a. Remove 3 fasteners.

11. Position aside oil filter drain (left side of engine).

    a. Remove 2 fasteners.

    b. Position oil drain filter aside.

12. Remove ground wire attached to the left engine bracket.

13. Position aside starter wire harness bracket.

14. Remove left hand engine support bracket.

    a. Remove the 3 fasteners.

15. **4x2 Non-EDGE:** Loosen all powertrain fasteners (transmission insulator/retainer and engine mounts) and go to Step 20.

NOTE
SOME FASTENERS WILL REMAIN LOOSE TO ALLOW FOR NEUTRALIZATION OF THE MOUNTS AND BRACKETS, LATER IN THIS PROCEDURE.

4x4 perform Steps a-e

a. Disconnect the axle vent.

b. Support the axle with a suitable jack.

c. Remove the left front axle to frame bolt and nut (Figure 2).

d. Remove the other two axle to frame bolts and nuts (Figure 3).

e. Lower the front axle.

All 4x4 and Edge 4x2

16. Remove engine support insulator nuts, both sides (Figure 4).

17. Remove motor mounts.

18. Position the new mounts (left-side 2L5Z-6038-AA and right-side 2L5Z-6038-AB) and install nuts. 4x4 only-Tighten the bottom center nut. Torque to 51 ± 8 Lb-ft. (70 +/-10.5 N•m.

NOTE
LEAVE THE TOP TWO NUTS AND BOTTOM CENTER NUT ON THE 4X2 LOOSE, FOR EACH MOTOR MOUNT.

All 4x4 perform Steps a-e

a. Raise the front axle assembly into position.

b. Install axle to frame bolts and two nuts.

NOTE
USE NEW FASTENERS WHEN INSTALLING THE FRONT AXLE (FIGURE 5).

c. Install the left front axle to frame bolt and nut (Figure 6).

d. Position the jack aside.

e. Connect the axle vent tube.

All 4x4 and Edge 4x2

19. Remove the old transmission insulator and replace with new transmission insulator/retainer (2L5Z-6068-AA). Refer to section 307-1.

NOTE
LEAVE THE FASTENERS LOOSE.

All Vehicles

NOTE
THIS STEP CAN BE PERFORMED THROUGH THE FENDER WELL.

20. For All 4x4 and EDGE 4x2, attach the new engine support brackets (left- side 2L5Z-6031-AA and right-side 2L5Z-6030-AB) to the engine (3 bolts). Torque to 51 ± 8 Lb-ft. (70 +/-10.5 N•m. For 4x2 non-EDGE, attach the new engine support brackets (left-side 2L5Z-6031-DA and right-side 2L5Z-6030-DA) to the engine (3 bolts). Torque to 51 ± 8 Lb-ft. (70 +/-10.5 N•m.

21. Reposition Starter Wire Harness Bracket.

22. Install ground wire to the left engine support bracket. Torque to 89 Lb-in (10 N•m).

23. Position back oil filter drain funnel. Torque to 89 Lb-in. (10 N•m).

24. Lower vehicle.

25. Lower the engine and remove the three-bar engine support.

26. Install the engine support bracket nuts to the frame loosely (Figure 7).

27. Remove the engine lifting eyes.

28. Install the fan blade. Refer to section 303-03.

29. Install the hood.

30. Connect battery ground. Refer to Section 414-01.
31. Move the vehicle in forward and reverse 2-4 feet (.6-1.2 meters).
32. Raise and support the vehicle.
33. **All vehicles:** Tighten remaining engine mount, engine support bracket and transmission insulator/retainer fasteners. Torque to 51 ± 8 Lb-ft. (70 +/-10.5 N•m.
34. Lower vehicle.

PART NUMBER	PART NAME
2L5Z-6031-DA	Support Bracket-Left - 4x2 Non-EDGE
2L5Z-6030-DA	Support Bracket-Right - 4x2 Non-EDGE
2L5Z-6031-AA	Support Bracket-Left - All 4x4 And EDGE 4x2
2L5Z-6030-AB	Support Bracket-Right - All 4x4 And EDGE 4x2
2L5Z-6038-AA	Engine Mount - Left
2L5Z-6038-AB	Engine Mount - Right
2L5Z-6068-AA	Transmission Insulator and Retainer

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: Eligible Under The Provisions Of Bumper To Bumper Warranty Coverage

OPERATION	DESCRIPTION	TIME
022204A	Replace Engine Mount Brackets And Perform Powertrain Neutralization Procedure (Includes Time To Perform WDS Cylinder Power Balance Test) Ranger 4X2 Non Edge.	2.1 Hrs.
022204B	Replace Engine Mount Brackets, Engine Mounts And Transmission Insulator Then Perform Powertrain Neutralization Procedure (Includes Time To Perform WDS Cylinder Power Balance Test) Ranger 4X2 Edge.	2.8 Hrs.
022204C	Replace Engine Mount Brackets, Engine Mounts And Transmission Insulator Then Perform Powertrain Neutralization Procedure (Includes Time To Perform WDS Cylinder Power Balance Test) Ranger 4X4.	3.0 Hrs.

DEALER CODING

BASIC PART NO.	CONDITION CODE
6046	07

OASIS CODES: 499000, 608000, 608400, 703000, 703100

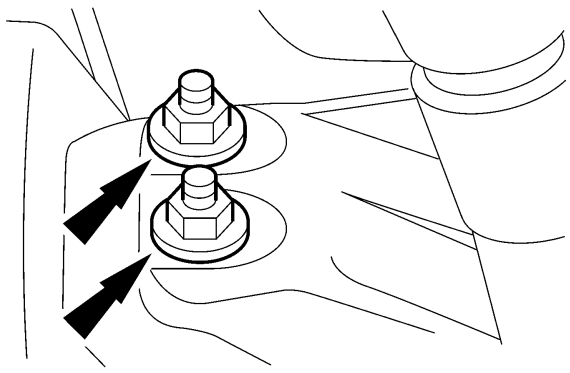


Figure 1 - Article 02-22-4

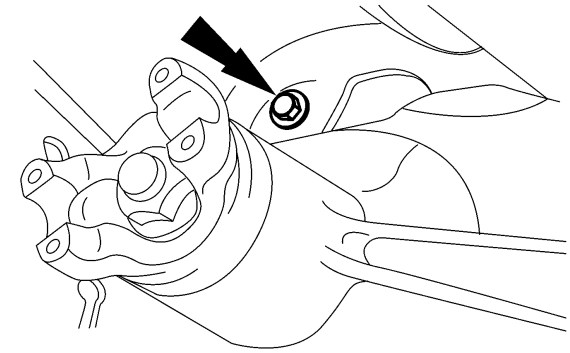


Figure 2 - Article 02-22-4

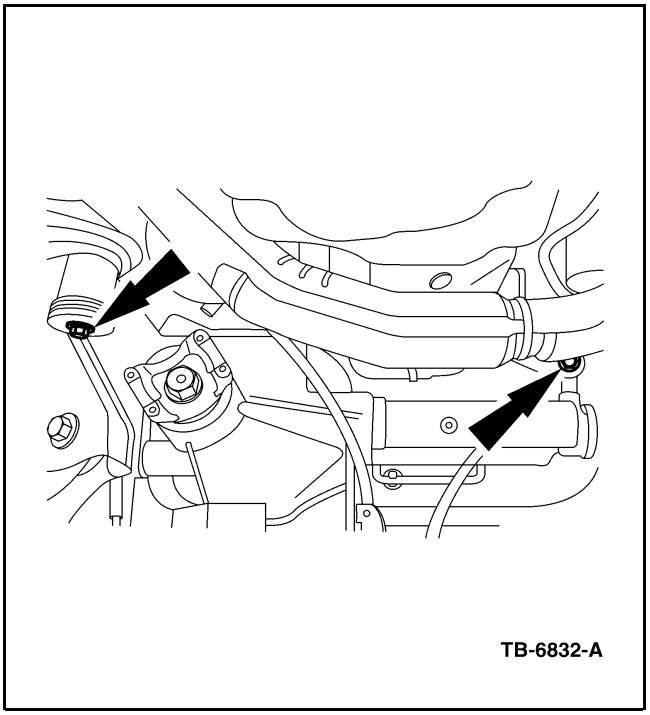


Figure 3 - Article 02-22-4

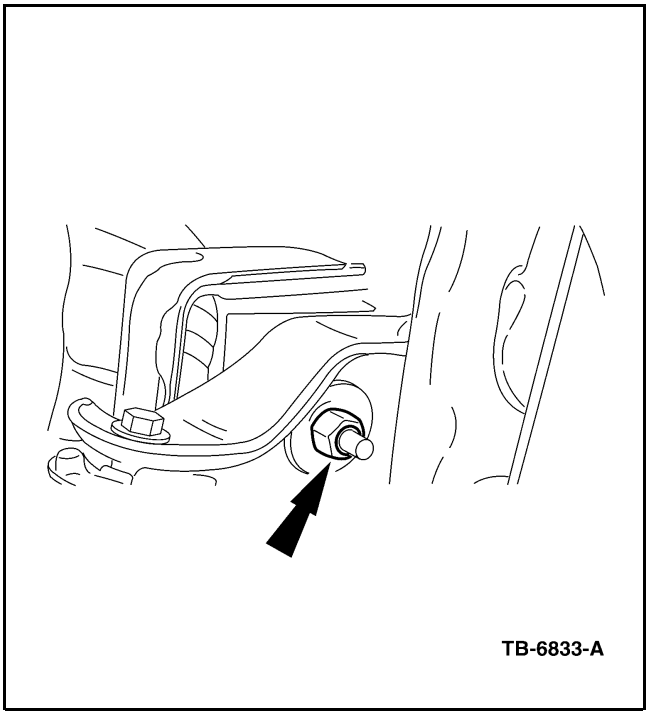


Figure 4 - Article 02-22-4

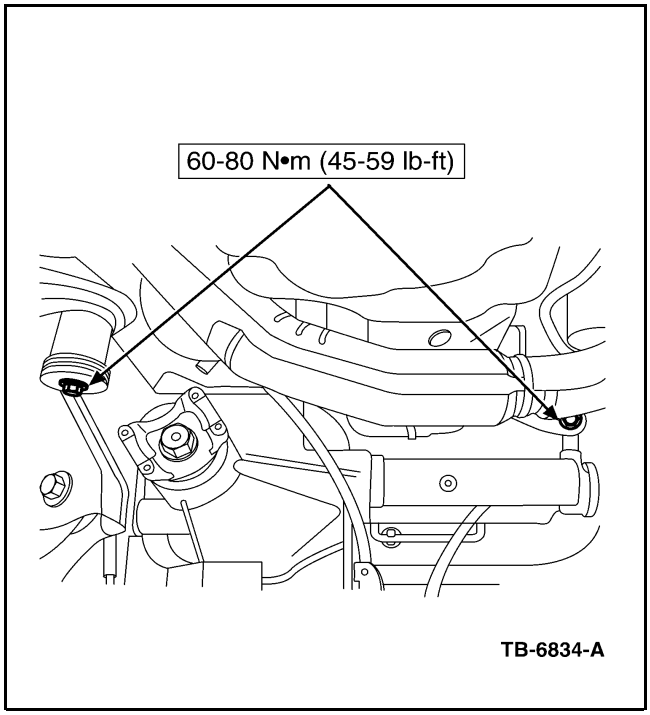


Figure 5 - Article 02-22-4

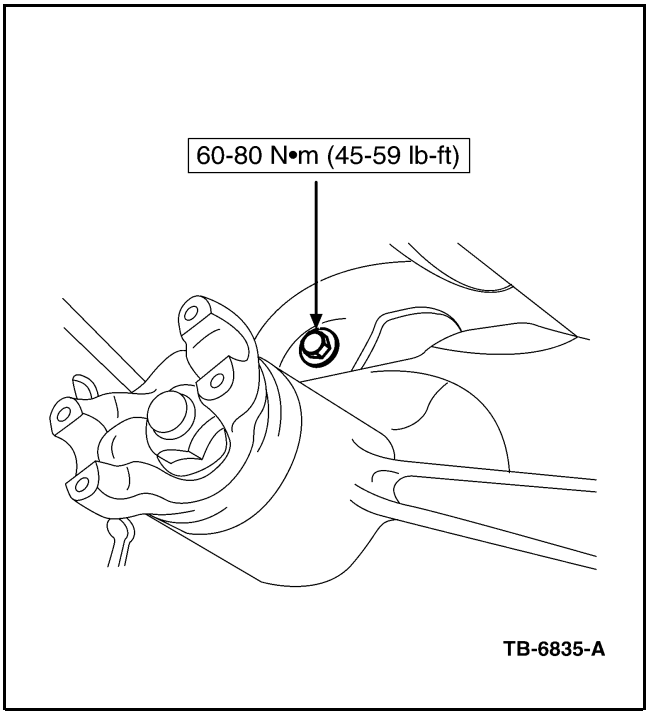


Figure 6 - Article 02-22-4

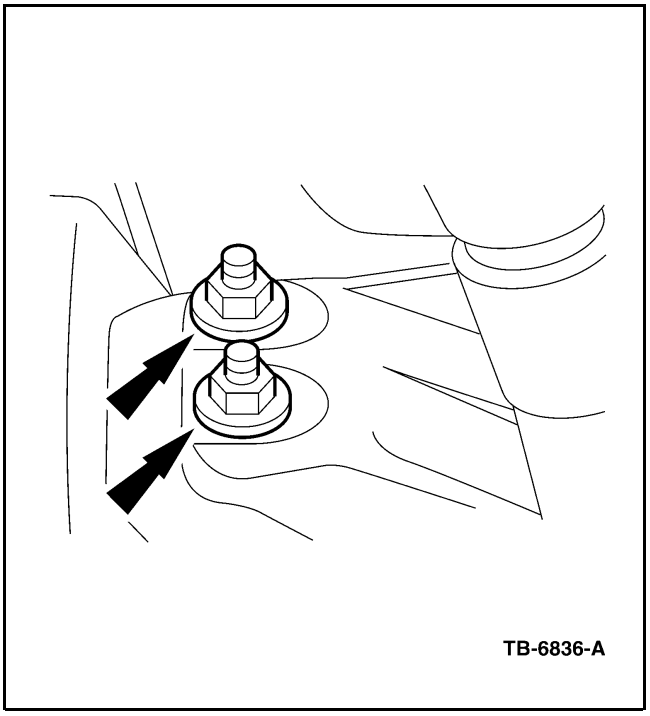


Figure 7 - Article 02-22-4

NOTE: The information in Technical Service Bulletins is intended for use by trained, professional technicians with the knowledge, tools, and equipment to do the job properly and safely. It informs these technicians of conditions that may occur on some vehicles, or provides information that could assist in proper vehicle service. The procedures should not be performed by "do-it-yourselfers". Do not assume that a condition described affects your car or truck. Contact a Ford, Lincoln, or Mercury dealership to determine whether the Bulletin applies to your vehicle.